Manufacturer

Example Forge Works GmbH

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Customer

Example Steel Trading GmbH

Musterstrasse 10
70173Stuttgart
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Digital Material Passport

ID	DMP-TEST-MULTIROW-001	*Version*	1.0.0
Issue Date	2025-04-10	*Certificate Type*	EN 10204 3.1

Business Transaction

Order		**Delivery**	
Order ID	ORD-2025-3312	*Delivery ID*	DLV-2025-4455
Quantity	18000 kg	*Quantity*	18000 kg

Product Information

Product Name	Forged round bars
Batch ID	BATCH-2025-03-7832
Surface Condition	Rough turned
Delivery Condition	+N - (Normalized)

Product Norms

Standard	EN 10250-1
Standard	EN 10250-2

Material Designations

Name (EN 10027-1)	S355J2 +N
Number (EN 10027-2)	1.0577 +N

Product Shape

Form	RoundBar
Length	2800 mm
Diameter	350 mm

Packaging and Marking

Marking
Both ends stamped with grade and heat number.

Packaging
Bundled, max 20 MT

Coloring
blue

Heat Treatment

Process	*Lot*	*Furnace*	*Date*
Normalizing	HT-LOT-2025-0315-B	FURNACE-07	2025-03-15

Stages

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Austenitizing	910 C			Air

Chemical Analysis

Heat Number	H-2025-7832
Melting Process	EAF+LF+VD
Casting Method	IngotCasting

Elements

Symbol	**C**	**Si**	**Mn**	**P**	**S**	**Cr**	**Ni**	**Mo**	**Cu**
Unit	%	%	%	%	%	%	%	%	%
Min	0.1	-	1.1	-	0.015	-	-	-	-
Max	0.22	0.55	1.65	0.03	0.03	0.35	0.35	0.1	0.3
Actual	0.17	0.28	1.38	0.014	0.022	0.18	0.14	0.05	0.22

Symbol	**B**	**Al**	**V**	**Sn**	**Ti**	**Sb**	**Ti**	**Sb**	**F1**
Unit	%	%	%	%	%	%	ppm	%	%
Min	-	0.015	-	-	-	-	-	-	-
Max	-	0.06	0.025	0.015	0.006	0.003	3.0	0.5	0.52
Actual	4.0E-4	0.028	0.012	0.006	0.004	0.0012	2.1	0.38	0.37

Formula Definitions

F1 = Cr+Ni+Mo: 0.37%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength in delivery condition Specimen: 1/2T, L - ID: TS-001	Rm	548N/mm²	440N/mm²	630N/mm²	EN ISO 6892-1/B1	-
Yield Strength in delivery condition Specimen: 1/2T, L - ID: YS-001	Re	372N/mm²	320N/mm²		EN ISO 6892-1	-
Elongation in delivery condition Specimen: 1/2T, L - ID: EL-L-001	A	26%	17%		EN ISO 6892-1/B1	-
Elongation in delivery condition Specimen: 1/2T, T - ID: EL-T-001	A	20%	14%		EN ISO 6892-1/B1	-
Reduction of area in delivery condition Specimen: 1/2T, L - ID: RA-001	Z	48%			EN ISO 6892-1	-
Impact test ISO-V -20°C, in delivery condition Specimen: 1/2T, T - ID: IT-T-001					EN ISO 148-1	-

Individual Values	# 1	# 2	# 3
Value [J]	36	30	37

Statistics	Mean	Min/Max	Std Dev
	34.33	22	

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Impact test ISO-V -20°C, in delivery condition Specimen: 1/2T, L - ID: IT-L-001					EN ISO 148-1	-

Individual Values	# 1	# 2	# 3
Value [J]	62	65	58

Statistics	Mean	Min/Max	Std Dev
	61.67	38	

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Surface hardness		162HB		180HB		-

Physical Properties

Property	Symbol	Actual	Target/Min	Maximum	Method	Status
Grain size		6	4	9	ASTM E112	-
Purity K4(ox)		20	-	35	DIN 50602	-
Purity K4(s)		25	-	35	DIN 50602	-
Straightness		1.2mm/m	-	2.0mm/m		-
Reduction ratio		3.8	3.0	-		-
Residual magnetism		5A/cm	-	10A/cm		-
fully formed structure, - no residue of dendritic - cast structure		Yes	-	-		-

Supplementary Tests

Property	Actual	Target/Min	Maximum	Method	Status
Ultrasonic test 100%	Yes Pass	-	-	EN 10228-3 class 3	-
Surface test	Yes 100% visually inspected, surface free from defects	-	-		-
Radioactivity	Yes max 100 Bq/kg Bq/kg	-	-		-

Validation

We hereby certify that the material described above has been manufactured, tested and inspected in accordance with the requirements of the applicable specifications and standards, and that the results comply with the specified requirements.

Individual Statements

✓ Material is of non-Russian origin *(EU Regulation No. 833/2014)*

Validated By

Name	*Title*	*Department*	*Date*
Dr. Maria Fischer	Head of Quality Assurance	Quality Assurance	2025-04-10